

Hangyu Yao

Birmingham, AL | Open to Relocate | Available: July 15, 2026 | Graduation: May 8, 2026 | 618-796-8666 | yaohangyu1@gmail.com | github.com/HangyuYao

PROFESSIONAL SUMMARY

M.S. Computer Science student focused on backend, software, and full-stack development with hands-on experience in Python, JavaScript, SQL, Docker, Git/GitHub, Linux, and web projects. Comfortable building and debugging customer-facing websites and data-driven software while bringing AI/ML experience in LLMs, PyTorch, computer vision, model evaluation, and AI-assisted coding with OpenAI Codex.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C++, MATLAB

Backend & Databases: SQL databases, REST API fundamentals, data processing, Docker, Linux, CRUD-style workflows

Frontend & Web: React fundamentals, HTML, CSS, responsive web design, Wix, Shopify theme customization

AI/ML: LLMs, deep learning, CNNs, PyTorch, scikit-learn, pandas, NumPy, matplotlib, model evaluation

Tools & Workflow: Git/GitHub, VS Code, Jupyter/Colab, OpenAI Codex, ChatGPT-assisted debugging, technical documentation

EXPERIENCE & SOFTWARE PROJECTS

Independent Web Developer - Client Web Development Projects

Top to Bottom Fishing Charter Website - Wix Web Design Project

2025

- Redesigned and updated a Wix website for a Panama City Beach fishing charter business, improving service pages, FAQ structure, navigation flow, contact/booking-oriented content, and customer-facing page layouts.
- Organized charter information, trip categories, business details, and common customer questions into clearer sections, improving visual consistency and making the site easier for visitors to understand and act on.

Wholesale Jewelry Website - Shopify Web Development Project

May 2026 - Present

- Currently building and customizing a Shopify wholesale jewelry website with category-based navigation, collection structure, and product catalog pages for necklaces, earrings, rings, bracelets, brooches, and mixed bulk products.
- Editing Shopify theme sections, templates, page layouts, navigation menus, and CSS to separate homepage content from catalog pages and improve storefront organization, visual consistency, and product browsing.
- Using OpenAI Codex and ChatGPT-assisted workflows to plan implementation options, debug Liquid/CSS layout issues, compare design alternatives, and review changes before applying final edits.

Aquatic Invertebrate Counts Using AI - Senior Design Project

2024

- Built a Python-based image analysis system inspired by YOLO to count and classify aquatic invertebrates from images, combining preprocessing, object-detection-style logic, and species identification workflows.
- Used Docker to containerize the project environment and Git/GitHub for source control, code sharing, and team collaboration across the senior design development cycle.
- Framed the project as an applied software system for automating a manual biological counting task, focusing on usability, reproducibility, and reliable image-based analysis rather than only model experimentation.

Network Protocol Server - Computer Networks Course Project

2023

- Built a command-driven client/server networking project with protocol-style request parsing, response codes, file-backed data storage, and article/group retrieval workflows.
- Tested local and remote retrieval paths while practicing backend concepts such as state management, error handling, synchronization logic, and network debugging.

AI / ML PROJECTS

LLM Chatbot Battle: User Preference Prediction - Deep Learning Course Project

2025

- Built machine learning and transformer-based models to predict preferred chatbot responses from human-labeled Chatbot Arena data, handling preprocessing, feature preparation, experimentation, and evaluation.
- Compared Logistic Regression, LightGBM, BERT, and Gemma using accuracy, F1-score, log-loss, ROC-AUC, and confusion matrices; explored LoRA and low-bit quantization for efficient fine-tuning.

Evaluating Leaf and Disease Segmentation Strategies for Plant Disease Classification - Master Project

2026

- Implemented PyTorch segmentation/classification experiments for plant disease analysis across multiple species and disease categories, comparing MobileNet, EfficientNet, and MCFFA-based architectures.
- Designed ablation studies to evaluate how leaf/lesion segmentation affected classification accuracy and inference efficiency, producing technical results for graduate research reporting.

EDUCATION

Southern Illinois University Edwardsville - M.S. in Computer Science, GPA: 3.62

Aug 2024 - May 8, 2026

Relevant Coursework: Advanced Artificial Intelligence, Image Processing, Advanced Algorithms, Database and Web Systems, Networking, Programming Languages

Southern Illinois University Edwardsville - B.S. in Computer Science, Minor in Mathematics, GPA: 3.279

Jan 2020 - May 2024